

Classifying Yetter-Drinfeld Modules over the Pansera Algebras H_{2n^2}

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Crash Course in Hopf Algebras

Algebras and Coalgebras

- An algebra is a vector space A with a ring structure:
 - Has *multiplication* $\mu : A \otimes A \rightarrow A$, *unit* $\eta : \mathbb{k} \rightarrow A$ satisfying:

$$\begin{array}{ccc} A \otimes A \otimes A & \xrightarrow{\text{id} \otimes \mu} & A \otimes A \\ \mu \otimes \text{id} \downarrow & & \downarrow \mu \\ A \otimes A & \xrightarrow{\mu} & A \end{array}$$

$$\begin{array}{ccccc} & & A \otimes A & & \\ & \nearrow \eta \otimes \text{id} & \downarrow \mu & \nwarrow \text{id} \otimes \eta & \\ \mathbb{k} \otimes A & & & & A \otimes \mathbb{k} \\ & \searrow \cong & \downarrow & \swarrow \cong & \\ & & A & & \end{array}$$

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 \end{array}$$

- A coalgebra is "dual" to an algebra, reversing the maps:
 - Has *comultiplication* $\Delta : C \rightarrow C \otimes C$, *counit* $\varepsilon : C \rightarrow \mathbb{k}$ satisfying:

$$\begin{array}{ccc}
 C \otimes C \otimes C & \xleftarrow{\text{id} \otimes \Delta} & C \otimes C \\
 \Delta \otimes \text{id} \uparrow & & \uparrow \Delta \\
 C \otimes C & \xleftarrow{\Delta} & C
 \end{array}$$

$$\begin{array}{ccccc}
 & & C \otimes C & & \\
 & \nwarrow \varepsilon \otimes \text{id} & \uparrow \Delta & \nearrow \text{id} \otimes \varepsilon & \\
 \mathbb{k} \otimes C & & & & C \otimes \mathbb{k} \\
 & \nwarrow \cong & \uparrow \cong & \nearrow \cong & \\
 & & C & &
 \end{array}$$

- Think of Δ as a splitting map, ε as a "projection" map.

- When we have a v.s H with an algebra structure μ, η and a coalgebra structure Δ, ε , we want some compatibility between them.

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- When the maps Δ, ε are algebra maps, we say H is a **bialgebra**.
- An *antipode* map $S : H \rightarrow H$ is a map satisfying:

$$\begin{array}{ccc} H & \xrightarrow{\Delta} & H \otimes H \\ \eta \circ \varepsilon \downarrow & & \downarrow S \otimes \text{id}_H \\ H & \xleftarrow{\mu} & H \otimes H \end{array} \qquad \begin{array}{ccc} H & \xrightarrow{\Delta} & H \otimes H \\ \eta \circ \varepsilon \downarrow & & \downarrow \text{id}_H \otimes S \\ H & \xleftarrow{\mu} & H \otimes H \end{array}$$

- Kind of an inverse map!

Hopf Algebras

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- Kind of an inverse map!
- When a bialgebra H has an antipode S , we say H is a **Hopf algebra**.

Examples of Hopf Algebras 1

- Let G be a finite group.
- Group Algebra: Take $\mathbb{k}G = \{\sum k_i g_i \mid k_i \in \mathbb{k}, g_i \in G\}$.
 - Multiplication by $\mu(g_1, g_2) = g_1 g_2$, extended linearly.
 - Comultiplication by $\Delta(g) = g \otimes g$, extended linearly.
 - Unit by $\eta(1) = e$, counit by $\varepsilon(g) = 1$, both extended linearly.
 - Antipode by $S(g) = g^{-1}$.
- Let's check how the antipode works:

$$\begin{array}{ccc}
 H & \xrightarrow{\Delta} & H \otimes H \\
 \eta \circ \varepsilon \downarrow & & \downarrow S \otimes \text{id}_H \\
 H & \xleftarrow{\mu} & H \otimes H
 \end{array}$$

- Split, invert one side, then multiply to remove all non-trivial information.

$$\begin{aligned}
 g &\xrightarrow{\Delta} g \otimes g \xrightarrow{S \otimes \text{id}} g^{-1} \otimes g \xrightarrow{\mu} e \\
 g &\xrightarrow{\varepsilon} 1 \xrightarrow{\eta} e
 \end{aligned}$$

Examples of Hopf Algebras 2

- Dual of a Group Algebra: Take: $(\mathbb{k}G)^* = \{f : \mathbb{k}G \rightarrow \mathbb{k} \mid \text{is linear}\}$.
 - This has a basis by $f_g : \mathbb{k}G$ defined by $f_g(h) = \delta_{gh}$.
 - Multiplication by $\mu(f_{g_1}, f_{g_2})(h) = f_{g_1}(h)f_{g_2}(h)$.
 - Comultiplication by $\Delta(f_g)(h) = \sum_{\substack{g_1, g_2 \in G \\ g_1 g_2 = g}} f_{g_1} \otimes f_{g_2}$.
 - Unit by $1 \rightarrow \text{id}$, counit by $f_g \rightarrow \delta_{ge}$, both extended linearly.
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 - Antipode by $f_g \rightarrow f_{g^{-1}}$.
- Important: Dual Group Algebra has a commutative multiplication.
- General Hopf algebras can be thought of as generalizing the algebra of functions on a group with a non-commutative multiplication.

- The Pansera algebras H_{2n^2} are a very interesting Hopf algebra.
 - Small enough to work with easily.
 - Still exhibits lots of interesting behavior.
 - Good "toy" example to understand general phenomena.

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 - Small enough to work with easily.
 - Still exhibits lots of interesting behavior.
 - Good "toy" example to understand general phenomena.
- Essentially one copy of $\mathbb{k}[\mathbb{Z}_n \times \mathbb{Z}_n]$ and another "twisted" copy.

Modules, Comodules, Yetter-Drinfeld Modules

Modules and Comodules

- An A -module is a v.s V with a *action* $\rho : A \otimes V \rightarrow V$ satisfying:

$$\begin{array}{ccc}
 A \otimes A \otimes V & \xrightarrow{\mu \otimes \text{id}_V} & A \otimes V \\
 \text{id}_A \otimes \rho \downarrow & & \downarrow \rho \\
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- A C -comodule is a v.s V with coaction $\delta : V \rightarrow C \otimes V$ satisfying:

$$\begin{array}{ccc}
 V & \xrightarrow{\delta} & C \otimes V \\
 \delta \downarrow & & \downarrow \Delta \otimes \text{id}_V \\
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- We can think of a coaction as a factoring map, sending

$$v \rightarrow \sum_i c_i \otimes v_i$$

Yetter-Drinfeld Modules

- What happens when we have V both a module and a comodule?
- When these satisfy a certain compatibility condition, we say we have a *Yetter-Drinfeld module*.

Yetter-Drinfeld Modules

- What happens when we have V both a module and a comodule?
- When these satisfy a certain compatibility condition, we say we have a *Yetter-Drinfeld module*.
- Why do we care about these?
 - Encodes "combinatorial" information about the Hopf algebra.
 - The Andruskiewitsch–Schneider program provides a method for classifying finite-dimensional Hopf algebras over H_{2n^2} .
 - Analogue of classifying groups with a given quotient group.
 - First ingredient in above method is the Yetter-Drinfeld Modules.

Problem and Results

Question: What are the simple Hopf algebras over H_{2n^2} ?

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- Hopf algebras are important tools.
 - Lie algebras encode infinitesimal symmetry.
 - The q -deformation of a Lie algebra is a Hopf algebra, and represents quantum infinitesimal symmetry.
 - Concrete applications in quantum computing, TQFTs.

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- Hopf algebras are important tools.
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 - The q -deformation of a Lie algebra is a Hopf algebra, and represents quantum infinitesimal symmetry.
 - Concrete applications in quantum computing, TQFTs.
- Classification of Hopf algebras is a fundamental question.
 - Unfortunately, little is known.
 - Classifying Hopf algebras over H_{2n^2} already advances classification.
 - Insights gained could be used in more general classification programs.

Our Results

Theorem (2025, Prasad, Pollastri, Plavnik, Spencer)

The following is a complete list of simple comodules over H_{2n^2} :

- 1. There are n^2 one-dimensional simple comodules.*
- 2. There is one n -dimensional simple comodule.*

In particular, the category of comodules over H_{2n^2} can be realized as a Tambara-Yamagami category.

Theorem (2025, Prasad, Pollastri, Plavnik, Spencer)

The following is a complete list of simple Yetter-Drinfeld modules over H_{2n^2} :

- 1. There are $2n^2$ 1-dimensional simple Yetter-Drinfeld modules.*
- 2. There are $\frac{n^2(n^2-1)}{2}$ 2-dimensional simple Yetter-Drinfeld modules.*
- 3. There are $2n^2$ n -dimensional simple Yetter-Drinfeld modules.*

In particular, the category of Yetter-Drinfeld modules over H_{2n^2} can be realized as the center of a Tambara-Yamagami category.

Future work includes:

- Finishing analysis of Yetter-Drinfeld Modules
- Computing the fusion rules of ${}^{H_{2n^2}}_{H_{2n^2}}\mathcal{YD}$
- Applying this to determine quasitriangular structures over H_{2n^2}
- Use this to find the Nichols algebras over H_{2n^2}
- Ultimately classify Hopf algebras over H_{2n^2} .

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 - Many thanks to Dr. Anderson Trimm and Dr. Micah Fogel at the Illinois Mathematics and Science Academy for teaching me most of the foundational math I know.
 - Many thanks to my parents and friends for supporting me throughout the research process.

THANK YOU FOR LISTENING!
Questions?



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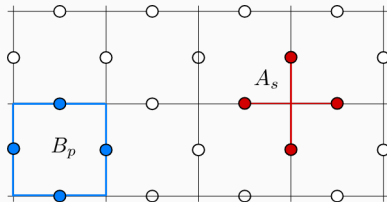
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Kitaev's Toric Code

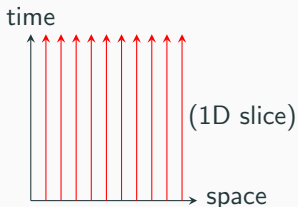
- Embed $\mathbb{Z}^2$ on a torus (with periodic boundary conditions).



- A, B operators act on various qubits, and are stabilizers.
- A qubit lies on each lattice. Anyonic states (electric and magnetic) are excitations of states.
- The anyons of the toric code are exactly the irreducible objects of the category $\text{Rep}(D(\mathbb{Z}_2)) \cong {}_{\mathbb{Z}_2}^{\mathbb{Z}_2}\mathcal{YD}$.
- Braiding rules, controlled by the quasitriangular structure of $D(\mathbb{Z}_2)$, control quantum error.
- Can be done with any Hopf algebra ${}^H_H\mathcal{YD}$.

Topological Quantum Field Theories

- 1. Reshetikhin–Turaev takes Ribbon Hopf Algebras, produces a TQFT. This gives lots of interesting examples of TQFTs.
- 2. In physics, we want to compute the *partition function*, which describes statistical-physical information about a TQFT.
- We compute along 1-dimensional slices:



- When combining these slices, quantum non-commutativity demands the input of solutions to the Yang-Baxter equation.
- These arise from R -matrices, which are closely related to braidings.